Comparison of well-orderings: given any two, one is isomorphic to an initial segment of the other

The Open Logic Project contributors

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-set-theory-ordinals/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the Separation definition of f, its isomorphism property with the named citation, the initial-segment closures, and the least-elements contradiction are all verbatim in the span.

Editorial note. The label thm:woalwayscomparable names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own thm environment in the unit source.

Allowed standard prerequisites (editorial).

- Order comparison of ordinal-segment isomorphisms. Same unit, \ollabel{lemordsegments} (source line 820). At line 854: pairs of isomorphic initial segments are ordered coherently, so the assembled relation f is an isomorphism between its domain and range.
- Initial segments and ordinal membership. Same unit, \ollabel{wellordinitialsegment} (source line 799). At line 859: downward closure of dom(f) and ran(f) inside the two well-orders.
- Set formation by Separation. The unit's standing ZF background; invoked in-line at 847 'Using Separation'. Carves the comparison relation f out of the product A x B at lines 848-852.

Source reference. Original lemma lemordsegments: sources/olp-set-theory-ordinals/pinned-original-source.tex, source lines 820–820. Prerequisite pointer invoked at proof line(s) 854; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference wellordinitialsegment: sources/olp-set-theory-ordinals/pinned-original-source.tex, source lines 799–799. Prerequisite pointer

invoked at proof line(s) 859; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Theorem 0.1. *Given any two well-orderings, one is isomorphic to an initial segment (not necessarily proper) of the other.*

Proof. Let $\langle A, <\rangle$ and $\langle B, \lessdot\rangle$ be well-orderings. Using Separation, let

$$f = \{\, \langle a, b\rangle \in A \times B \mid \langle A_a, <_a\rangle = \langle B_b, \lessdot_b\rangle \,\}.$$

By lemordsegments, $a_1 < a_2$ iff $b_1 \lessdot b_2$ for all $\langle a_1, b_1\rangle, \langle a_2, b_2\rangle \in f$. So $f\colon \operatorname{dom} f \to \operatorname{ran} f$ is an isomorphism.

If $a_2 \in \operatorname{dom} f$ and $a_1 < a_2$, then $a_1 \in \operatorname{dom} f$ by wellordinitialsegment; so $\operatorname{dom} f$ is an initial segment of A. Similarly, $\operatorname{ran} f$ is an initial segment of B. For reductio, suppose both are *proper* initial segments. Then let a be the $<$-least !!element of $A \setminus \operatorname{dom} f$, so that $\operatorname{dom} f = A_a$, and let b be the $\lessdot$-least !!element of $B \setminus \operatorname{ran} f$, so that $\operatorname{ran} f = B_b$. So $f\colon A_a \to B_b$ is an isomorphism, and hence $\langle a, b\rangle \in f$, a contradiction. $\square$